Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 22)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
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TOTAL	

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Answer all questions in the spaces provided.

0 1 An aquifer beneath an agricultural plain holds 3.6 x 10^{11} m^3 of water. It is recharged at 1.2 x 10^9 m^3 per year, and 4.8 x 10^9 m^3 per year is abstracted from it.

0 1 . 1 Residence time = volume of the reservoir / rate of flow through it. Calculate the residence time of the water in this aquifer, using the recharge rate.

[2 marks]

Answer _____ years

0 1 . 2 Explain the properties a rock must have to form a useful aquifer.

[3 marks]

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0 1 . 3 Explain why the water table beneath the plain is falling.

[2 marks]

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0 1 . 4 Describe **four** consequences of continuing to over-abstract this aquifer.

[4 marks]

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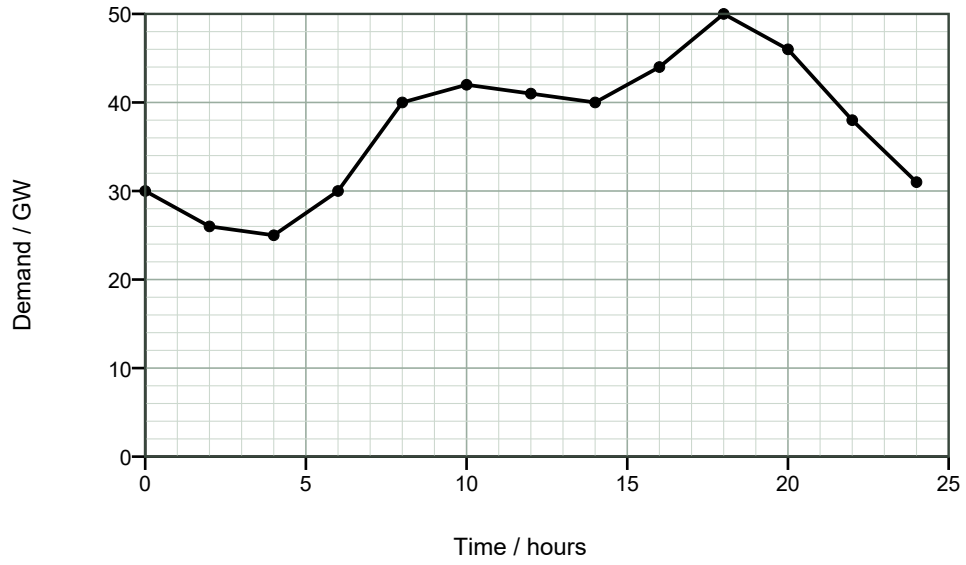
0 1 . 5 Explain why an aquifer holding 'fossil water' cannot be exploited sustainably, and describe **two** measures that would reduce abstraction.

[4 marks]

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0 2 Figure 1 shows the demand for electricity in the UK over a 24-hour period on a weekday in winter.

Figure 1



0 2 . 1 Describe the pattern of demand shown in Figure 1.

[2 marks]

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0 2 . 2 Explain the causes of the fluctuations shown in Figure 1 and give **two** other causes of fluctuations in electricity demand.

[4 marks]

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0 2 . 3 Explain why base-load power stations (nuclear, large coal) produce a surplus of electricity at night.

[2 marks]

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0 2 . 4 Give **two** causes of fluctuations in energy **supply**.

[2 marks]

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0 3 . 1 Which of the following best describes the **precautionary principle**? Tick **one** box.

[1 mark]

A A waste is assumed to cause pollution until research shows it is unlikely to do so ☐

B The polluter must pay for any damage caused ☐

C Emissions should be as low as reasonably achievable ☐

D Only pollutants known to be toxic need to be controlled ☐

0 3 . 2 Explain why the polluter pays principle works best when penalties are high and enforcement is reliable.

[2 marks]

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0 3 . 3 Give **two** choices about emission location that reduce the severity of pollution.

[2 marks]

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0 4 Table 1 shows the water quality required for four different uses.

0 4 . 1 Complete Table 1 by giving the quality requirement for each water use and the reason for it.

[4 marks]

Table 1

Water use	Quality requirement	Reason	
Potable (public) supply			
Spray irrigation			
Power station condenser cooling			
Industrial boiler water			

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0 4 . 2 Explain why aeration is used in the treatment of water from anaerobic sources.

[2 marks]

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0 4 . 3 Explain how flocculation removes clay particles that sedimentation cannot.

[2 marks]

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0 4 . 4 Give **two** methods used to sterilise treated water.

[2 marks]

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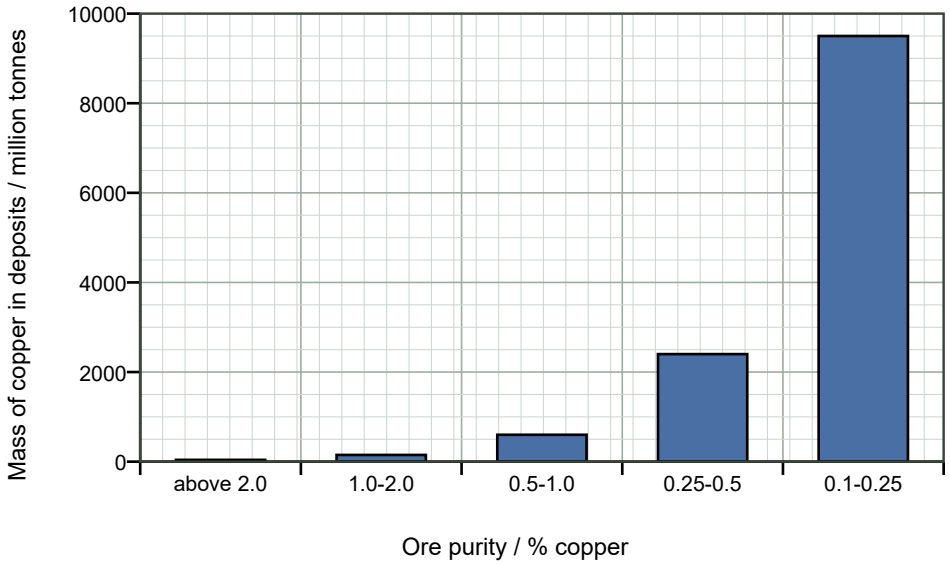
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0 5 Figure 1 shows the mass of copper contained in the Earth's crust at different ore purities, illustrating Lasky's principle.

Figure 1



0 5 . 1 State Lasky's principle.

[1 mark]

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0 5 . 2 Use Figure 1 to calculate how many times more copper is contained in deposits of 0.25-0.5% purity than in deposits above 2.0%.

[1 mark]

Answer ____ times

0 5 . 3 Explain the difference between the **reserves** and the **resource** of a mineral.

[3 marks]

0 5 . 4 Explain, using Lasky's principle, why the ability to exploit slightly lower grade ores greatly increases reserves.

[2 marks]

0 5 . 5 Explain why the main problem for future mineral supplies is not the quantity that exists in the crust.

[3 marks]

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0 6 . 1 Which of the following oil spill treatments is most likely to kill organisms living deep in beach sediments that were not affected by the oil? Tick one box.

[1 mark]

A Absorbent mops ☐

B Steam washing ☐

C Inflatable booms ☐

D Polymerising agents ☐

0 6 . 2 Explain how polymerising agents make an oil spill easier to clean up.

[2 marks]

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0 6 . 3 Explain why deoxygenation is a more common problem in water than in air.

[2 marks]

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0 7 Table 1 shows global reactive nitrogen fixed each year by natural processes and by human activities.

Table 1

Source	Nitrogen fixed / million tonnes per year
Biological fixation (natural ecosystems)	110
Lightning	5
Haber process (fertiliser manufacture)	120
Cultivation of legume crops	60
Combustion (NOx)	30

0 7 . 1 Use Table 1 to calculate the percentage of total nitrogen fixation that is caused by human activities.

[2 marks]

Answer ____ %

0 7 . 2 Describe the Haber process and explain why it is unsustainable compared with biological nitrogen fixation.

[3 marks]

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0 7 . 3 Describe **three** consequences of the increased movement of nitrogen compounds caused by human activities.

[3 marks]

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0 7 . 4 Explain how composting or anaerobic digestion of biological wastes produces a better fertiliser than raw waste.

[2 marks]

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0 8 . 1 Complete Table 1 by naming the most suitable statistical test for each hypothesis.

[4 marks]

Table 1

Hypothesis	Statistical test	
Noise levels decrease as distance from a road increases		
Fewer cars use a road after road charging is introduced (counts on 12 days before and after)		
Woodlands managed in different ways have different numbers of dormice		
Mean lobster mass is higher in areas where collection is banned		

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0 8 . 2 Explain why the chi-squared test cannot be used to compare the dissolved oxygen concentrations of two rivers.

[2 marks]

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0 8 . 3 Explain why a scientific investigation never 'proves' a hypothesis.

[2 marks]

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0 8 . 4 Describe how error bars are added to a graph of means and how they are interpreted.

[2 marks]

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0 9 Table 1 shows the per capita energy consumption and GDP per capita of five countries.

Table 1

Country	GDP per capita / US$ thousand	Energy use per capita / GJ per year	Main energy resource
A	62	290	Oil, gas
B	44	125	Gas, nuclear, renewables
C	10	95	Coal
D	2	25	Biomass (wood), oil
E	55	220	Hydroelectricity, oil

0 9 . 1 Describe the relationship between GDP per capita and energy use per capita shown in Table 1.

[2 marks]

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0 9 . 2 Explain why affluence increases per capita energy consumption.

[3 marks]

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0 9 . 3 Suggest **two** reasons why country B uses much less energy per capita than country A despite a similar level of affluence.

[2 marks]

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0 9 . 4 Explain why the per capita energy consumption of the UK has fallen in recent years even though consumption of goods has risen.

[2 marks]

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0 9 . 5 Give **one** way energy is used in each of the following: water supplies; agriculture.

[1 mark]

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1 0 . 1 Explain how the properties of smoke and of the gases involved in acid rain lead to their environmental effects, and evaluate the effectiveness of the methods used to control them.

[9 marks]

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1 0 . 2 State **one** effect of tropospheric ozone on plants.

[1 mark]

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Write an essay on one of the following topics.

11 . 1 Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions.

[25 marks]

OR

11 . 2 Discuss how the properties of solar power and hydroelectric power affect their use, and evaluate how new technologies could increase their contribution to energy supplies.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS